

BOM

Component	Mass (g/cell)	kg/kWh	Source annotation
Positive electrode active material (NMC811)	16.168	0.8895	coating mass x 0.96 (lit. default split)
Positive conductive additive (carbon black)	0.337	0.0185	coating mass x 0.02 (lit. default)
Positive binder (PVDF)	0.337	0.0185	coating mass x 0.02 (lit. default)
Negative electrode active material (graphite)	9.692	0.5332	coating mass x 0.955 (lit. default split)
Negative conductive additive (carbon black)	0.152	0.0084	coating mass x 0.015 (lit. default)
Negative binder (SBR+CMC)	0.304	0.0167	coating mass x 0.03 (lit. default)
Separator (PE, 12um, porosity 0.47)	0.259	0.0143	calc-energy layer mass (1-porosity) x density x area
Electrolyte (LiPF6-based, advanced formulation)	6.966	0.3832	pore volume 5.81 cm ³ x 1.2 g/cm ³ (lit. density, annotated)
Positive current collector (Al foil 16um)	4.437	0.2441	calc-energy layer mass
Negative current collector (Cu foil 12um)	11.042	0.6075	calc-energy layer mass
Enclosure and tabs	0	0	Not modeled (parameter set lacks cell-can mass)
TOTAL (incl. electrolyte, excl. enclosure)	49.696	2.7339	cell energy 18.178 Wh (sim integration)
Enclosure/tabs	Not modeled	Not modeled	parameter set lacks cell-can mass